22/09/26 Experiment - 8/7 : NMap to discover live hosts using NMap scans or TryHackMe platform

Aim : To discover live hosts using NMap scans (ARP, TCMP, TCP/UDP) on TryHackMe platform

Procedure :-

Task 1 : Introduction

(*) ARP Scan : This scan ~~wants~~ wants ARP requests to discover live hosts.

(*) ICMP Scan : To identify live hosts.

(*) TCP / UDP Ping scan : Sends packets to TCP ports & UDP ports to determine live hosts.

Task 2 : Subnetworks

Network Segment - It is collection of computers linked together through common medium.

Subnetwork : It is usually the equivalent of one or more network segments linked together or configured to utilize the same router in an IP networks.

Subnets with /16 or /24.

Task 3

(i) List :-

* <Machine-IP> <scanme.nmap.org>

(ii) Range :-

* 10.11.12.15-20
* 5 scan 6 IP addresses.

(iii) Subnet :-

<machine-IP>/30

Nmap -SL <Targets>

Task 4

Discovering Live Hosts :-

* ARP for man in middle attack.
* ICMP from network layer used in DDO's attacks also UDP.
* TCP from transport layer for SYN flooding.

Task 5 : NMap host discovery using ARP.

(i) When privileged user scans NMap uses ARP requests.

(ii) ~~NMap -PR~~

nmap -PR -sn <Targets>

(iii) nmap -PR -sn <machine IP> /24

(iv) nmap -PR -sn <attackbox> /24

Task 6 : NMap host discovery using ICMP

* nmap -PE -sn Target to check host time
* -PP option inputs nmap to ICMP timestamp requests.

Task 7 : NMap host discovery using TCP/UDP

* nmap -PS -sn Target checks TCP port is open or not.
* -PS21 , -PS25 targets posts 21, 22, 23, 24, 25.
* TCP ACK Ping run as privileged user.
* UDP Ping - Sends UDP packets.
* nmap -PU -sn Target

Task 8: Using reverse-DNS lookup

* NMap reverses-DNS internet hosts.
* -R option to query the DNS server over offline hosts.

Task 9: Final Task

ARP scan, ICMP echo scan, ICMP timestamp scan, ICMP address mask scan, TCP SYN Ping

Result: Thus the experiment was carried out successfully.